

ASSIGNMENT #2

33. PSO – CRM System

Software Scope and Effort Estimation

Pakistan State Oil Company Limited
Customer Relationship Management System

Group Members

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TASK (a) – Part 1: Function Point Analysis

This section performs a complete Unadjusted Function Point (UFP) count for the PSO CRM system. Function Points are categorised across five standard IFPUG components. For each component, the number of Data Element Types (DETs) and Record Element Types (RETs) are explicitly justified through reference to the abstract system design established in Assignment 1.

2.1 External Inputs (EI)

External Inputs represent all data entry transactions that cross the system boundary into the PSO CRM. The FP complexity thresholds for EI are: Low (≤ 4 DETs or 1 RET $\rightarrow$ 3 FP), Average (5–15 DETs, 2–3 RETs $\rightarrow$ 4 FP), and High (> 15 DETs or > 3 RETs $\rightarrow$ 6 FP).

Req. ID	EI Name	Data Elements (DETs)	File Types (RETs)	DET Count	Complexity	FP Weight
FR01	Customer Registration	CNIC, Full Name, Mobile Number, Vehicle Registration No., Vehicle Make/Model, Customer Type (Fleet/Individual)	1 RET – Customer Profile	7	Low	3
FR02	POS Offline Transaction Queuing	Liters Purchased, Fuel Type, Transaction Amount, POS Terminal ID, Timestamp, Offline Sync Flag, Cashier ID	2 RETs – Transaction Log + Sync Queue	7	Average	4
FR03	Batch Data Synchronisation	Batch ID, Transaction List Array, Checksum Total, Station ID, Network Status, Sync Timestamp, Error Flags	2 RETs – Batch Record + Station Record	7	Average	4
FR04	Points Calculation & Awarding	Purchase Amount, Tier Multiplier, Points Awarded, Customer ID, Current Balance, New Balance, Tier Status	2 RETs – Customer Profile + Loyalty Tier	7	Average	4
FR05	Point Redemption Authorisation	One-Time Password (OTP), Redemption Amount, POS ID, Customer ID, Updated Points Balance, Redemption Log Entry	1 RET – Redemption Log	6	Low	3

EI Sub-total Calculation: $(2 \text{ Low} \times 3 \text{ FP}) + (3 \text{ Average} \times 4 \text{ FP}) = 6 + 12 = 18 \text{ FP}$

2.2 External Outputs (EO)

External Outputs are data or control information transmitted outside the system boundary, involving derived data or system-generated reports. The FP complexity thresholds for EO are: Low (≤ 5 DETs, 0–1 RETs $\rightarrow$ 4 FP), Average (6–19 DETs, 2–3 RETs $\rightarrow$ 5 FP), and High (> 19 DETs or > 3 RETs $\rightarrow$ 7 FP).

Req. ID	EO Name	Data Elements (DETs)	File Types (RETs)	DET Count	Complexity	FP Weight
FR06	Transaction SMS Notification	Customer Name, Points Awarded/Redeemed, Current Balance, Station Name, Fuel Type, Date/Time	2 RETs – Customer Profile + Transaction	6	Low	4
FR07	Fleet Manager Analytics Dashboard	Total Fuel Consumed, Points Earned, Vehicle Breakdown, Station Heatmap, Date Range Filter, Cost Centre Code, Export Flag	3 RETs – Customer + Vehicle + Transaction	7	Average	5

EO Sub-total Calculation: (1 Low × 4 FP) + (1 Average × 5 FP) = 4 + 5 = 9 FP

2.3 Internal Logical Files (ILF)

Internal Logical Files are user-identifiable groups of logically related data that are maintained within the system boundary. The FP complexity thresholds for ILF are: Low (1 RET, 1–19 DETs → 7 FP), Average (2–5 RETs, 1–19 DETs → 10 FP), and High (>5 RETs or >19 DETs → 15 FP).

Req. ID	ILF Name	Record Types (RETs)	RET Count	DET Description	Complexity	FP
FR08	Unified Customer Database	Customer Profile, Vehicle Log, Transaction History, Points Balance, Loyalty Tier	4	Full Name, CNIC, Mobile, Email, Vehicle Make, Reg. No., Fuel Type, Litres Purchased, Transaction Amount, Points Earned, Points Redeemed, Tier Status, Registration Date, Last Activity Date, Sync Status (15 DETs)	Average	10

ILF Sub-total Calculation: (1 Average × 10 FP) = 10 FP

2.4 External Interface Files (EIF)

External Interface Files are user-identifiable groups of logically related data referenced by the system but maintained by an external application. The SAP ERP is a wholly external system managed outside the PSO CRM boundary.

Req. ID	EIF Name	External Source	RET Count	DET Description	Complexity	FP
FR09	SAP/ERP Financial Ledger	PSO Legacy SAP ERP System	2	Account ID, Daily Point Liability (PKR), Monetary Equivalent Value, Station Reconciliation Flag, Sync Timestamp (5 DETs)	Low	5

EIF Sub-total Calculation: (1 Low × 5 FP) = 5 FP

2.5 External Inquiries (EQ)

External Inquiries are input/output combinations that retrieve data from the system without producing derived data or updating any ILF. They are read-only query transactions. The FP complexity thresholds for EQ are: Low (1–5 DETs, 0–1 RETs → 3 FP), Average (6–19 DETs, 2–3 RETs → 4 FP).

Req. ID	EQ Name	Files Referenced	RET Count	DET Description	Complexity	FP
FR10	Real-time Balance Inquiry	Unified Customer Database	1	Customer ID, CNIC Number, Current Points Balance, Equivalent Rupee Value (4 DETs)	Low	3

EQ Sub-total Calculation: $(1 \text{ Low} \times 3 \text{ FP}) = 3 \text{ FP}$

2.6 UFP Summary

The following table consolidates all five function point components to produce the final Unadjusted Function Point (UFP) count for the PSO CRM system.

FP Component	Requirements Mapped	Count	Complexity Calculation	FP Count	Cumulative FP
External Inputs (EI)	FR01, FR02, FR03, FR04, FR05	5	Low: $2 \times 3 = 6$ Avg: $3 \times 4 = 12$	18	18
External Outputs (EO)	FR06, FR07	2	Low: $1 \times 4 = 4$ Avg: $1 \times 5 = 5$	9	27
Internal Logical Files (ILF)	FR08	1	Average: $1 \times 10 = 10$	10	37
External Interface Files (EIF)	FR09	1	Low: $1 \times 5 = 5$	5	42
External Inquiries (EQ)	FR10	1	Low: $1 \times 3 = 3$	3	45
TOTAL UFP	FR01 – FR10	10	—	45	45

TOTAL UNADJUSTED FUNCTION POINTS (UFP): 45 FP | 10 Requirements | Spread across 5 IFPUG Component Types

TASK (a) – Part 2: Functional and Non-Functional Requirements

The following requirements were used as the basis for the Function Point Analysis above. Each functional requirement maps to a specific FP component type, and each non-functional requirement constrains the quality attributes of the system's implementation.

3.1 Functional Requirements (FR01 – FR10)

The ten functional requirements below define the core behavioural capabilities of the PSO CRM system. Each requirement is categorised by its FP type and accompanied by a detailed description of its intended behaviour within the real-world PSO operating environment.

#	Requirement	FP Type	Description
FR01	Customer Registration	EI	Fuel station managers register new customer profiles via POS terminal using CNIC, mobile number, vehicle registration details, and classification (individual or fleet). The system validates CNIC uniqueness and activates the loyalty account.
FR02	POS Offline Transaction Queuing	EI	When internet connectivity is lost at a fuel station, the local POS system logs each loyalty transaction to an on-device queue, flagging it with a sync-pending status for later upload when connectivity is restored.
FR03	Batch Data Synchronisation	EI	Upon network restoration, the POS client module automatically compresses and pushes all queued offline transactions to the central CRM server in a verified batch, with checksum validation to ensure data integrity.
FR04	Points Calculation & Awarding	EI	The Loyalty Engine reads the purchase amount and the customer's current tier multiplier to calculate DigiCash points earned. The customer balance is immediately updated and the new tier re-evaluated.
FR05	Point Redemption Authorisation	EI	The system sends a One-Time Password (OTP) to the registered mobile number. Upon verification, the system authorises a fuel discount equivalent to the redeemed points and logs the transaction.
FR06	Transaction SMS Notification	EO	The CRM automatically triggers an SMS via a third-party SMS gateway (e.g., TeleCommunication Ltd.) immediately after each transaction, detailing points earned or redeemed and the updated current balance.
FR07	Fleet Manager Analytics Dashboard	EO	Corporate clients and fleet owners can access a secure web portal to view aggregated fuel consumption, points earned per vehicle, station-level heatmaps, cost-centre breakdowns, and exportable reports.
FR08	Unified Customer Database	ILF	The central repository at PSO HQ stores all customer demographic data, vehicle records, transaction histories, points balances, and loyalty tier statuses for all registered customers nationwide.
FR09	SAP/ERP Financial Ledger Integration	EIF	The CRM reads and writes aggregated daily point liabilities to PSO's existing SAP ERP backend, ensuring all DigiCash point balances are reflected in the company's financial accounts for regulatory compliance.
FR10	Real-time Balance Inquiry	EQ	Station attendants or customers can query the current points balance using a CNIC number without executing any transaction, enabling fast, read-only access to loyalty account status.

3.2 Non-Functional Requirements (NFR01 – NFR10)

The following non-functional requirements define the quality constraints and operational standards under which the PSO CRM system must operate. They are directly informed by the unique challenges of Pakistan's oil and gas sector, including remote station connectivity, regulatory oversight, and legacy infrastructure compatibility.

#	Requirement	Category	Description
NFR01	Offline Availability	Reliability	The POS module must retain up to 72 hours of offline transaction data without any data loss, using encrypted local SQLite storage, to handle prolonged internet outages at remote stations.
NFR02	Horizontal Scalability	Scalability	The CRM backend must support concurrent data synchronisation from all 3,500+ PSO retail outlets simultaneously without degradation, supporting auto-scaling Kubernetes pod deployment.
NFR03	Transaction Response Time	Performance	Point redemption OTP verification and balance update queries must resolve end-to-end in under 2.5 seconds to prevent queuing at fuel pumps during peak hours.
NFR04	PII Data Encryption	Security	All CNIC numbers, mobile numbers, and financial transaction data must be encrypted at rest (AES-256) and in transit (TLS 1.3), with field-level tokenisation for CNIC values in the database.
NFR05	Legacy Hardware Compatibility	Usability	The local POS sync agent must operate within 512 MB RAM and run on Pentium-class CPUs, allowing deployment on legacy hardware at rural stations without requiring hardware upgrades.
NFR06	Regulatory Compliance	Compliance	The system architecture, data retention policies, and audit logging must fully comply with State Bank of Pakistan (SBP) Payment Systems Regulations and SECP data governance directives.
NFR07	ERP Integration Uptime	Reliability	The SAP/ERP integration bridge must maintain 99.9% uptime (≤ 8.7 hours downtime/year) using an active-passive failover configuration for daily financial reconciliation processing.
NFR08	Bilingual Interface Support	Usability	All customer-facing outputs — including SMS alerts, POS receipts, and the customer mobile app — must render correctly in both Urdu (Naskh script) and English, driven by a customer language preference flag.
NFR09	Modular Independent Deployment	Maintainability	The Loyalty Engine microservice must be independently deployable and updatable using blue-green deployment without requiring downtime for the Customer Database or POS integration layers.
NFR10	Low-Bandwidth Synchronisation	Performance	Batch sync payloads must be compressed to under 20 KB per transaction using Protocol Buffers (protobuf) serialisation to ensure effective operation over 2G/3G networks at remote rural stations.

Development Duration	≈ 3.07 Months	12.30 PM ÷ 4 persons
Adjusted Timeline (incl. testing/UAT)	12 Months	Aligned with business case rollout plan

This COCOMO-II analysis confirms that the PSO CRM project, with a total UFP of 45, a mixed-technology stack, and moderate platform difficulty driven by legacy hardware and low-bandwidth connectivity constraints, is a well-scoped, medium-complexity enterprise project. The estimated 12.30 Person-Months of core development effort is consistent with industry benchmarks for comparable CRM deployments in the energy and utilities sector, and provides PSO's project management office with a defensible, data-driven basis for budgeting, team assembly, and schedule planning.